

1. Statement of the Art

1.1 SAE Interpretability

Sparse Autoencoders (SAEs) have become an important tool in large-model interpretability research and have played a significant role in tasks such as steering, causal intervention, and model behavior analysis. Therefore, improving the decomposition capability of SAEs over internal model representations, as well as the quality of the features they learn, is of substantial practical value.

The core idea of an SAE is to decompose highly superposed and difficult-to-interpret Transformer activations into a set of sparse latent features, thereby extracting representational structures that are closer to monosemanticity and more semantically readable. Steps are as follow:

1. The encoder maps features to a sparse latent variable space.

$$z = f_{en}(x) = \sigma(W_{en}x + b_{en})$$

2. By using a decoder, it's lossly reconstructed as $\hat{x}$.

$$\hat{x} = f_{de}(z) = W_{de}z + b_{de}$$

3. $\|x - \hat{x}\|_2^2$ is the reconstruction error, and $\lambda\|z\|_1$ is the sparsity penalty term, used to make latent activation as sparse as possible.

$$L_{SAE} = \|x - \hat{x}\|_2^2 + \lambda\|z\|_1$$

Since modern language models often encode multiple semantic factors within the same activation space, SAEs provide an unsupervised way to extract feature structures that are more amenable to human analysis from such entangled representations. Research on SAEs has developed along several directions, including stronger sparsity constraints, improved dictionary learning and feature selection mechanisms, and more comprehensive evaluation frameworks, leading to variants such as Top-K SAE, Gated SAE, and RouteSAE.

However, most existing work still focuses on post-hoc decomposition, that is, improving the SAE itself in order to better decompose representations that already exist in the model. In contrast, there is still a lack of systematic research on modifying the model architecture itself so that it produces representations that are more suitable for SAE decomposition. This study therefore focuses on this upstream representation design problem, with the goal of making Transformer representations more decomposable and interpretable.

1.2 what-where disentanglement

Transformers, in their pursuit of performance, often learn highly superposed internal representations. Although such representations enhance modeling capacity, they also increase entanglement among features, making post hoc decomposition and interpretability

analysis more difficult. This raises a central question: can we obtain representations that are easier to decompose after training without substantially compromising model performance?

Compared with vision tasks, semantic information in images is often tightly coupled with spatial location, so what cannot usually exist independently of where. For example, the same red pixels can correspond to very different semantics when they appear on an apple versus on blood. In contrast, the semantic identity of a token in LLMs is often relatively stable across positions—for instance, “apple” in “I eat an apple” and “apple” in “Apple is sweet” largely preserves its core meaning. This suggests that, in LLMs, content and position may be partially separable, providing a natural basis for what-where disentanglement.

Existing work on what-where disentanglement is already substantial, but it is often treated as a secondary issue within positional encoding design or SAE interpretability analysis. From this perspective, this paper aims to connect positional encoding with SAE interpretability through the lens of what-where disentanglement, and to explore whether reducing what-where entanglement can improve the decomposability and interpretability of SAE representations without sacrificing model performance.

1.3 Positional Encoding

Since the introduction of the Transformer, researchers have proposed a variety of PE methods, including sinusoidal PE, learned PE, T5-style relative biases, ALiBi, and RoPE. As noted earlier, language models tend to rely more on relative positional information to model contextual relationships.

ALiBi and RoPE use different absolute PE methods to represent relative position information. ALiBi adopts a **translational transformation**, introducing a linear distance-based bias into attention, whereas RoPE adopts a **rotational transformation**, applying rotations to the query and key vectors so that the attention scores primarily depend on relative positional differences. In this paper, we do not discuss ALiBi further and instead focus on RoPE, which has become the mainstream choice.

1. We get word embedding input x with d dimensions in position p .

$$x^p = [x_1^p, x_2^p, \dots, x_d^p]$$

2. Then, split it into 2D vector.

$$x^p = (x_1^p, x_2^p, x_3^p, \dots, x_d^p) \Rightarrow (x_{1:2}^p, x_{3:4}^p, \dots, x_{d-1:d}^p)$$

3. We apply Rotation matrix R_θ for every group. The i_{th} group is as follow:

$$\begin{pmatrix} q_{2i-1}^p \\ q_{2i}^p \end{pmatrix} = R_\theta \begin{pmatrix} x_{2i-1}^p \\ x_{2i}^p \end{pmatrix} = \begin{pmatrix} \cos(p\theta_i) & -\sin(p\theta_i) \\ \sin(p\theta_i) & \cos(p\theta_i) \end{pmatrix} \begin{pmatrix} x_{2i-1}^p \\ x_{2i}^p \end{pmatrix}$$

$$\theta_i = 10000^{-2(i-1)/d}$$

4. We make it complex form. $\hat{i}$ means imaginary number. The i_{th} pair of dimension is:

$$x_i^p = (x_{2i-1}^p, x_{2i}^p) \Rightarrow (\text{complex form}) C_i^p = x_{2i-1}^p + \hat{i}x_{2i}^p$$

5. According to Euler's formula: $e^{i\theta} = \cos\theta + \hat{i}\sin\theta$, we can represent the the after rotation i_{th} pair as a complex multiplication:

$$C_i^{p'} = C_i^p \cdot e^{ip\theta_i}$$

6. Matrix R_θ could be regarded as rotating degree θ in complex space. The rotation parameter in complex space is:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

7. Expanding formula in part 3, we can get:

$$q_{2i-1}^p + iq_{2i}^p = (x_{2i-1}^p + ix_{2i}^p) \cdot (\cos(p\theta_i) + i\sin(p\theta_i))$$

8. If complex $a + ib = c + id$, then $a = c, b = d$, so can we obtain the rotated components:

$$q_{2i-1}^p = x_{2i-1}^p \cos(p\theta_i) - x_{2i}^p \sin(p\theta_i)$$

$$q_{2i}^p = x_{2i}^p \sin(p\theta_i) + x_{2i-1}^p \cos(p\theta_i)$$

x_{2i}^p represent what, $p\theta_i$ represent where. While strong entanglement between what and where may improve representational capacity, it also weakens the decomposability of the learned representations, thereby making it difficult for SAE to recover sufficiently pure semantic features in the feature

PoPE was proposed to enhance the what-where disentanglement of RoPE. PoPE represents each scalar channel as a complex number in *polar form*, where the *magnitude* carries what and the *phase* carries where.

1. We get word embedding input x with d dimensions at position p :

$$x^p = [x_1^p, x_2^p, \dots, x_d^p]$$

2. Unlike RoPE, PoPE does not group every two dimensions into a 2D Cartesian vector.

Instead, it treats each scalar feature as one channel and converts it into the *magnitude* of a complex number:

$$x^p = [x_1^p, x_2^p, \dots, x_d^p]$$

3. PoPE maps each scalar channel into a non-negative magnitude using the softplus function $\sigma(x) = \ln(1 + e^x)$, which allows that value to be interpreted as a magnitude in polar coordinates. μ means magnitude.

$$\mu_i^p = \sigma(x_i^p) = \ln(1 + e^{x_i^p})$$

4. PoPE assigns position only to the *phase*. For the i_{th} channel, the phase is determined purely by the token position p and the channel frequency θ_i , so content no longer shifts the phase. ϕ means phase

$$\phi_i^p = p\theta_i$$

$$\theta_i = \theta^{(i-1)/d} \text{ (} \theta \text{ usually use 10000)}$$

5. Now each channel is written directly in polar-complex form. Its magnitude μ_i^p stores content, and its phase ϕ_i^p stores position

$$q_i^p = \mu_i^p e^{i\phi_i^p} = \mu_i^p e^{ip\theta_i}$$

6. According to Euler's formula, the complex number can be expanded into Cartesian coordinates, but unlike RoPE, this expansion starts from a magnitude-phase representation rather than from a 2D semantic pair

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$q_i^p = \mu_i^p (\cos(p\theta_i) + i\sin(p\theta_i))$$

If we write the real and imaginary parts explicitly, they are obtained from the same magnitude μ_i^p and the same position phase $p\theta_i$. This is the key structural difference from

RoPE: the semantic quantity enters only through the magnitude, not through the phase offset.

PoPE has shown promising performance on a range of position-sensitive tasks. Nevertheless, there is still a lack of systematic studies comparing PoPE and RoPE in the context of SAEs. Motivated by this gap, our work compares the two approaches from the perspective of SAE interpretability in an effort to fill this void.

I 2. Problem Definition

RQ1: Performance Stability

Does PoPE outperform RoPE in language model training and downstream tasks?

RQ2: Decoupling Effectiveness

Does PoPE achieve stronger content-position decoupling?

RQ3: Interpretability Improvement

Does PoPE enable SAEs to learn more monosemantic features?

RQ4: Mechanistic Impact

How does positional encoding affect attention logits' spectral structure (Toeplitz properties)?

Then to conclude:

Key Hypotheses to Test

1. Does what-where disentanglement harm language model training efficiency? If yes, how?
2. Does what-where disentanglement yield purer semantic features in SAE training? If yes, how?

This systematic comparison will clarify PoPE's practical value and inspire future positional encoding designs.

I 3. Achievements so far

Phase 1: Model Construction (Weeks 1-3) *Completed*

Limited by computing resources, we only train a clean 90M Toy Transformer from scratch with switchable RoPE/PoPE implementations to validate pipeline feasibility and test PoPE's positional processing hypothesis.

Remark: I planned to monkey patch Llama model and inject PoPE into it. But different PE will lead to very bad performance.

```
# Data Preparation
DATA_CFG = SimpleNamespace(
    dataset = "wikitext",
    config = "wikitext-103-raw-v1",
    key = "text",
    raw_text_samples = 30000,
```

```

train_blocks = 1500,
valid_blocks = 300,
seq_len = 1024,
tokenizer = "gpt2",
eos_join = False,
seed = 42,
)

DATA_RES = {
    vocab_size = 50257,
    total_token_used = 1891328,
    raw_text = 30000,
    non_empty_texts = 19357,
    used_texts = 14926,
}

```

```

# Model Construction
MODEL_CFG = SimpleNamespace(
    n_layers = 12,
    d_embed = 512,
    n_heads = 8,
    d_ff = 2048,
    dropout = 0.1,
    seq_len = DATA_CFG.seq_len,
    # Extrapolation test is not being considered at this time.
    lr = 3e-5,
    vocab_size = DATA_RES["vocab_size"]
)

MODEL_RES = {
    RoPE_para = 89343057, # about 90M
    PoPE_para = 89346129, # about 90M
}

```

Phase 2: Train & Evaluate (week3-6) *Completed*

- Stability Analysis: Compare training loss curves and gradient stability between the two architectures.
- Performance Evaluation: Benchmark models on language modeling (perplexity) and zero-shot length extrapolation tasks.

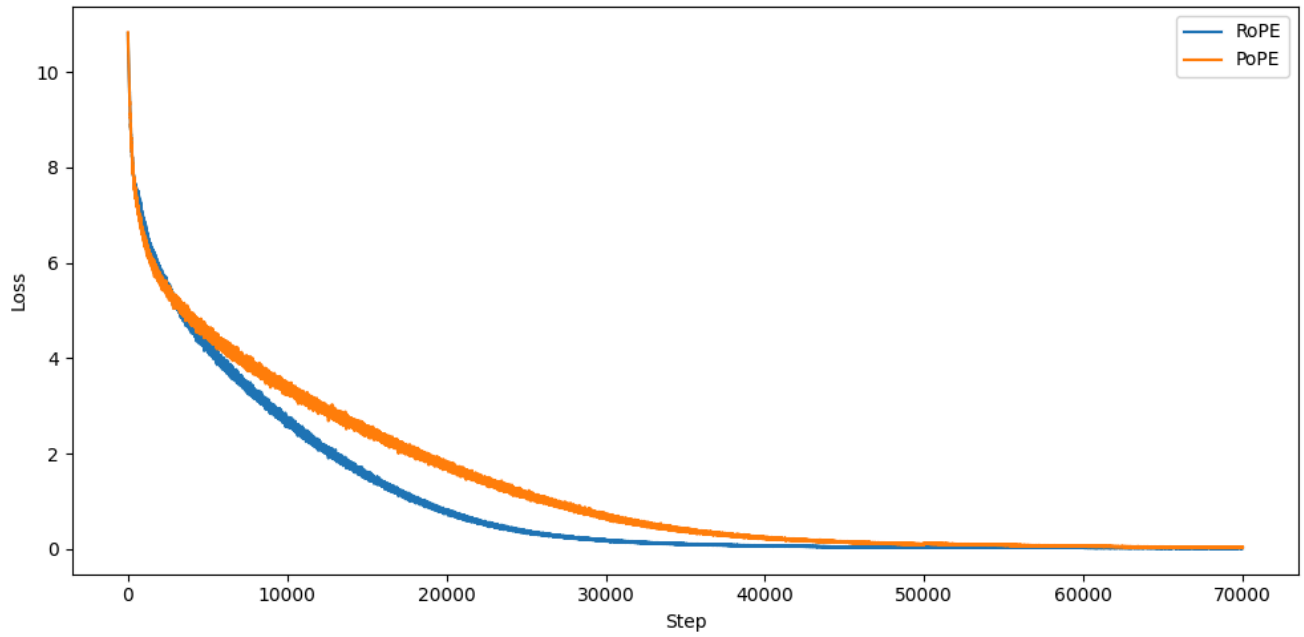
```

TRAIN = SimpleNamespace (
    steps = 70000,
    batch_size = 24,
    seq_len = MODEL.seq_len,
    save_interval = 10000,
    log_interval = 200,
)

```

```
data_type = "bfloat16",  
warmup = 2000  
)
```

Loss Curve:



model	layer	attn_entropy	avg_attn_distance
rope	0	5.83	243.78
pope	0	5.79	187.10
rope	1	2.83	238.01
pope	1	1.78	27.77
rope	2	4.12	205.83
pope	2	3.18	31.61
rope	3	4.91	189.53
pope	3	3.56	41.66
rope	4	4.97	172.63
pope	4	3.55	49.17
rope	5	4.75	147.42
pope	5	3.95	58.54
rope	6	4.67	132.54
pope	6	3.90	59.48
rope	7	4.35	116.70
pope	7	4.35	67.65
rope	8	4.69	119.71
pope	8	4.44	66.49
rope	9	4.63	109.58
pope	9	4.55	74.17
rope	10	4.50	95.96
pope	10	4.58	75.78
rope	11	4.63	108.91
pope	11	4.32	68.81

1. *attn_entropy*: Compute the Shannon entropy of each query's attention distribution by treating its normalized attention weights over all keys as a probability distribution. A lower entropy indicates that attention is more concentrated on a few positions, while a higher entropy indicates a more dispersed attention pattern.

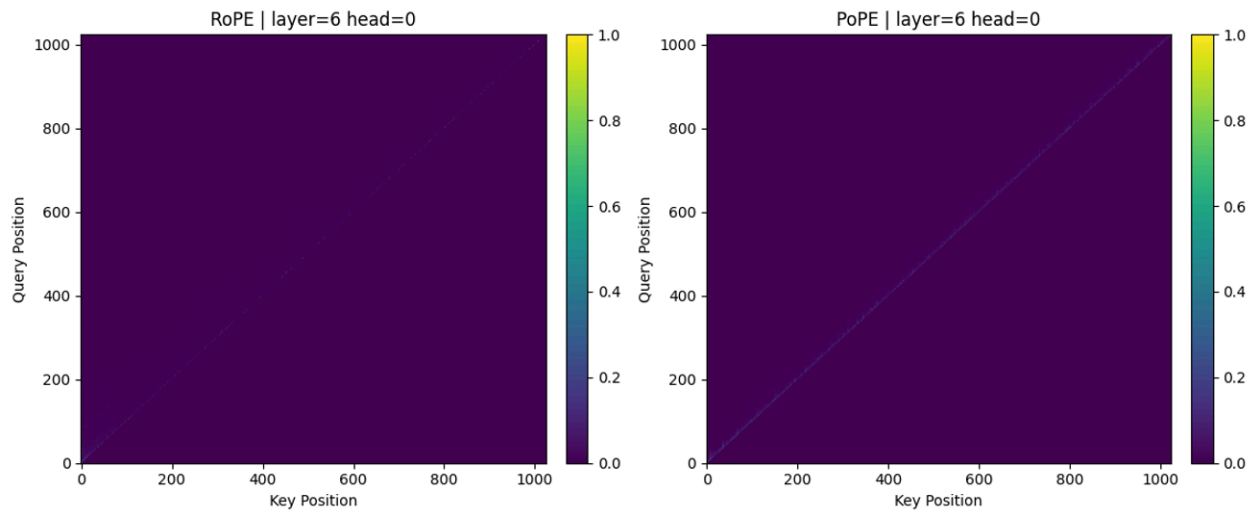
$H = -\sum_{i=1}^N a_i \log a_i$ is the attention weight at position i , N is the number of tokens paid attention to.

2. *avg_attn_distance*: Compute the attention-weighted average token distance, which reflects the model's average contextual span. A larger value indicates a stronger preference for long-range dependencies, whereas a smaller value indicates a stronger preference for local dependencies.

$Distance = \sum_{i=1}^N a_i d_i$

d_i is the distance between query word and the key word

3. *attn_heatmap*: High responses near the diagonal usually indicate that the model tends to focus more on adjacent or nearby tokens.



Phase 3: SAE Training & Core Metrics (Weeks 6-8) *Partly Completed*

Train SAEs (not top-k SAE) with identical configurations on matched RoPE/PoPE models across early/mid/late layers. Collect metrics as follow:

- Reconstruction Error
- L0 Norm
- Dead Features

```
SAE_CFG = SimpleNamespace(
    activation layer = "residual after block"
    dictionary size = 4096
    L1 coefficient = 1e-2
    epochs = 400
    learning rate = 2e-3
    batch size = 2048
    validation row cap = 8192 # sampling size of SAE validation set affects
    stability of MSE/nz.
)
```

Model	Val MSE	L0-Norm rate
RoPE	0.0261	8.59%
PoPE	0.0216	6.59%

From the current simple metrics, PoPE shows relatively low ValMSE and L0-Norm, which to some extent supports our hypothesis.

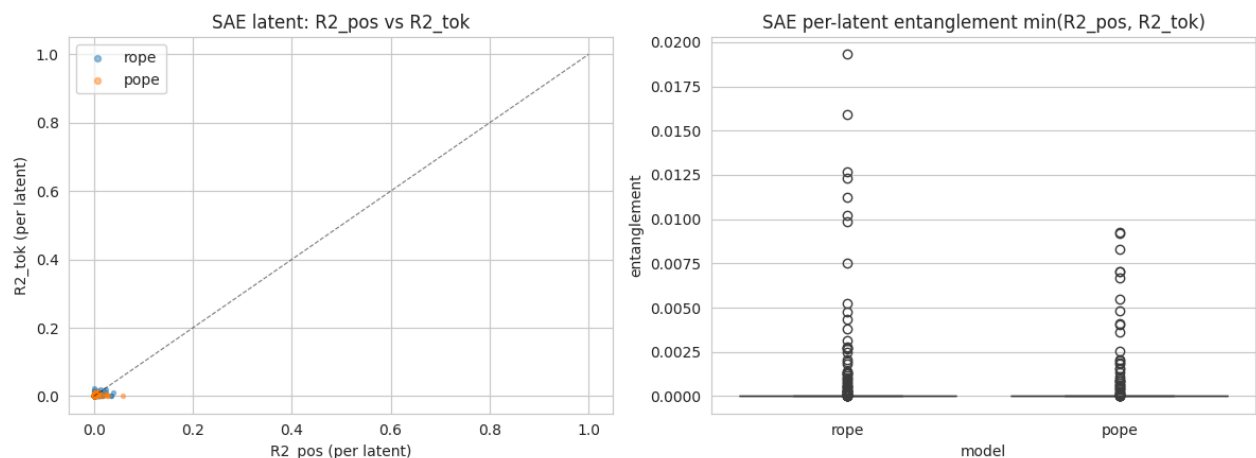
model	layer	mean_abs_corr	max_abs_corr	mean_mi	max_mi
rope	2	0.010157	0.997380	0.000437	0.136369
rope	6	0.006317	0.948309	0.000297	0.066435
rope	10	0.009525	0.998062	0.000383	0.053764
pope	2	0.013409	0.993653	0.000555	0.090393
pope	6	0.008905	0.937369	0.000400	0.039391
pope	10	0.007532	0.814278	0.000348	0.032652

- **Correlation:** measures the degree of linear relationship between features; lower values indicate weaker linear entanglement.
- **Mutual Information:** measures the mutual information between features; lower values indicate weaker nonlinear dependence and a greater degree of independence.

Phase 4: Disentanglement Benchmark (Weeks 9-11) *Partly Completed*

According to [Bilal Chughtai et al. \(2024\)](#), we use the Disentanglement Probe test, there are several important metrics as follow:

- R^2_{tok} : Indicates how single SAE feature could predict the semantic
- R^2_{pos} : Indicates how single SAE feature could predict the position



Phase 5: LLM-Assisted SAE Latent Interpretation (Weeks 11-13) *Partly Completed*

After each TopK-SAE training run, we use GPT-4o to automatically interpret all SAE latents and use the outputs as a quantitative measure of interpretability. Specifically, for each latent, we provide approximately 100–200 activation contexts together with their activation strengths and token positions, then use a fixed prompt to have GPT-4o generate a brief natural-language explanation of its activation pattern. Each latent is evaluated three times and the results are averaged to reduce randomness. The GPT-4o outputs are then used to assess the interpretation quality of each latent along the following three dimensions:

1. **Monosemanticity:** whether the latent corresponds to a single, clear concept.
2. **Low-level vs. high-level feature type:** whether the latent is more oriented toward low-level syntax, punctuation, or part-of-speech features versus high-level semantics, topics, or intents.

3. *Explanation examples*: whether the examples are representative enough to cover activation patterns across multiple different contexts.

Mainly refer work of [Paulo et al., \(2024\)](#) and [Shi et al., \(2025\)](#). Output samples are as follow:

```
gpt-4o interpret result:
```

```
AVG Monosemanticity Score: 3.95/5.00
```

```
AVG Low-level Features Score: 4.14/5.00
```

```
AVG High-level Features Score: 4.57/5.00
```

```
API Cost: $0.4261
```

```
Examples of Interpreted Features:
```

```
Feature #5: low-level (Score:5) - The activations are consistently associated with relative pronouns "which" and "that," indicating a ...
```

```
Feature #15: low-level (Score:4) - The activations are consistently associated with punctuation marks, particularly those used in dialo...
```

```
Feature #23: low-level (Score:4) - The activations are consistently associated with the token "than," which suggests a pattern related ...
```

I have also started using GPT-4o to help interpret features, but the code for a rigorous quantitative evaluation is not yet complete, so for now I will only present the following cases.

model	layer	summary	interpretation	evidence_span	feature_level
rope	2	Temporal or locational reference	The latent captures contexts where events or actions occur at specific times or places.	The highlighted token consistently appears in contexts indicating timing or location of actions.	high-level
rope	2	Definite article usage	This latent captures the strong presence of the definite article in various contexts.	The highlighted token consistently represents the definite article 'the', indicating its frequent and significant role in the contexts provided.	low-level
rope	6	Contextual relevance of specific tokens	The latent captures the significance of specific contextual tokens that denote key attributes or categories.	Tokens like 'young', 'unit', and 'conditions' indicate important descriptors or classifications in their contexts.	high-level

model	layer	summary	interpretation	evidence_span	feature_level
rope	6	punctuation and numerical formatting	This latent captures the activation related to punctuation and numerical representations in text.	The highlighted tokens consistently involve punctuation marks and numerical formats, indicating a focus on how numbers and symbols are structured in language.	low-level
rope	10	Prepositions indicating purpose or relation	The latent captures the use of prepositions that denote purpose, relation, or identity in various contexts.	The highlighted tokens consistently include prepositions like 'to', 'for', and 'of', which relate to actions or identities.	mixed
rope	10	Contextual relevance of key entities and actions	The latent captures the significance of key entities and actions within narrative contexts.	High activation is associated with important roles, relationships, and actions in the text.	high-level
pope	2	Numerical and temporal references	This latent captures contexts involving numerical values and time-related expressions.	The highlighted tokens consistently appear in contexts with numbers and time indicators.	high-level
pope	2	Past tense verb usage	The latent captures the use of past tense verbs indicating completed actions or states.	The highlighted tokens consistently represent past tense forms 'were' and 'was', showing a strong pattern of past actions or conditions.	low-level
pope	6	Temporal and relational markers	This latent captures the use of prepositions and conjunctions that indicate relationships and	The highlighted tokens often include prepositions and conjunctions that connect ideas or indicate time.	mixed

model	layer	summary	interpretation	evidence_span	feature_level
			temporal aspects in sentences.		
pope	6	Contextual reference to a cricket match incident	The latent captures references to a specific cricket match incident involving Mohammad Azharuddin's lbw decision.	All examples consistently refer to the same match incident with variations in context and phrasing.	high-level
pope	10	Definite articles in various contexts	The latent captures the prominence of definite articles in diverse textual contexts.	The highlighted tokens primarily include 'the' and 'a', indicating a focus on article usage.	mixed
pope	10	Contextual relevance of match events	This latent captures the context surrounding cricket match events, particularly focusing on the adjudication of lbw decisions.	All examples relate to a specific cricket match event involving Mohammad Azharuddin and the lbw decision, showing consistent context without activation.	high-level

I 4. Future Plan

Phase 6: Strengthening Experimental Reliability (Weeks 13-16)

1. Expand the experiments to larger datasets in contexts of varying complexity to test if what-where disentanglement will affect the learning ability in complex context?
2. Validate the results across multiple random seeds.
3. Introduce additional dynamic monitoring metrics to track representational changes, including:
 - SAE feature reuse rate
 - Monosemanticity growth
 - Dead feature evolution

Phase 7: Steering and Causal Intervention (Weeks 16-20)

The final phase evaluates the practical utility of the disentangled features. We will construct steering vectors from SAE features to intervene in model behavior, such as controlling

safety-related or stylistic outputs. We hypothesize that steering vectors derived from PoPE-based SAEs will achieve higher precision and fewer unintended side effects, due to the cleaner underlying feature space.

Phase 8: New Positional Embedding Design (Exploratory)

The insights from the preceding analyses may guide the design of a new positional embedding scheme that preserves strong representational capacity during learning while remaining more amenable to approximate decoupling in SAE analysis. Or combine different what-where extent, using different positional encoding for different head. (It's still just an imagination up to now; its feasibility hasn't been verified.)

I guess it may be helpful for multimodal tasks. But I need to read more and test more to verify my conjecture